AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational Databases

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15/ Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: 30
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choice questions [5 Marks]

This part consists of 5 questions carrying a weight of 1 mark for each.

1. Which of the following best describes a multivalued attribute in an ER diagram?
   a. An attribute with more than one name
   b. An attribute that can take multiple values for a single entity occurrence
   c. An attribute that is derived from other attributes
   d. An attribute that uniquely identifies the entity

2. In the conversion of a 1:N relationship from ERD to relational model, the foreign key is placed in:
   a. The entity on the "1" side
   b. The entity on the "N" side
   c. Both entities
   d. A separate new relation

3. Consider the ER schema:

   STUDENT(StudentID, Name)
   COURSE(CourseID, Title)
   ENROLLS(StudentID, CourseID, Grade)

   Which of the following statements is TRUE about its relational transformation?
   a. STUDENT and COURSE become relations, ENROLLS becomes a weak entity
   b. ENROLLS is a relationship and becomes a new relation with foreign keys referencing STUDENT and COURSE
   c. ENROLLS should be merged with STUDENT since it contains StudentID
   d. ENROLLS is redundant because COURSE already has CourseID

4. A total participation constraint means:
   a. Every entity occurrence must participate in at least one relationship occurrence
   b. Some entity occurrences may not participate in any relationship
   c. Every relationship must involve all entities in the schema
   d. An entity can only participate in one relationship at a time

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5. Which of the following is the correct transformation for a weak entity DEPENDENT(Name, Age) supported by EMPLOYEE(EmpID, Name) with partial key Name?
   a. DEPENDENT(DepID PK, Name, Age)
   b. DEPENDENT(Name PK, Age, EmpID FK)
   c. DEPENDENT(EmpID FK, Name PK, Age)
   d. DEPENDENT(EmpID PK, Name, Age)

Part 2: Problem solving questions [25 Marks]

Question 1 [8 Marks]

1- Consider the following ER model fragment:

- An **Athlete** can **PLAY** matches for a **Team**.
- An **Athlete** must also **BELONG TO** the same **Team**.
- An inclusion constraint (I) exists between PLAY and BELONG TO.

Explain how this inclusion constraint affects the relational schema. Write the relations and foreign keys needed. **[2 marks]**

2- In an ER model, an entity **PERSON** can be either a **STUDENT** or an **EMPLOYEE**. A **Totality constraint (T)** exists.
Explain what this constraint means in plain words and show two possible relational schema approaches to implement it. **[2 marks]**

3- Consider an entity **ITEM** that may either be a **PURCHASED ITEM** or a **PRODUCED ITEM**, but not both.

Which type of constraint is represented here? Show how this situation would be expressed in the relational model. What design issue may arise? **[2 marks]**

4- In an ER diagram, every **Order** must come **either** from a **Client** or from a **Service**, but not both.

Which type of constraint is modeled here? Translate this situation into a relational schema. What does the constraint guarantee? **[2 marks]**

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Question 2 [8 Marks]

Consider the following relational schema for a library system. For each relation, the attributes in **bold** constitute the key.

- Member(**MemberID**, MemberName, Age, Gender)
- Book(**BookID**, Title, Author, Genre)
- Loan(**#MemberID**, **#BookID**, LoanDate, ReturnDate)
- Librarian(**LibrarianID**, LibrarianName, Department)

Write the following queries in **relational algebra**:

1. What are the titles of books borrowed by the member "Alice"? **[2 marks]**
2. What are the names of members who borrowed books written by "J K Rowling"? **[2 marks]**
3. What are the titles of books **not** borrowed by any member? **[2 marks]**
4. What are the names of members who borrowed books from the genre "History" or "Philosophy"? **[2 marks]**

Question 3 [9 Marks]

A hospital needs a database to manage doctors, patients, and appointments.

- Each **Doctor** has a unique DoctorID, Name, and Specialty.
- Each **Patient** has a unique PatientID, Name, and DateOfBirth.
- A **Patient** can have multiple **Appointments**, but each Appointment is with exactly one **Doctor**.
- An Appointment has a unique AppointmentID, Date, and Reason.

1. Identify the entities, their attributes, and primary keys. **[3 marks]**
2. Draw the ER diagram representing the above scenario. **[3 marks]**
3. Convert the ER diagram into relational tables (schemas). **[3 marks]**

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End of questions

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